

Vessel-break sensitivity in a coupled 3D--1D model Completed controlled pilot, 11 September 2026.
All results are dimensionless and concern synthetic geometries.

Break location has a much larger effect than its small missing length alone suggests. For identical gaps removing 2.5% of a straight vessel, inlet-flux reductions are 68.39%, 29.20%, and 7.67% at 20%, 50%, and 80% of its length. A Y trunk break reduces inlet flux by 66.43%; a single daughter break reduces it by about 4.64%. In the larger tree, 32 breaks give a median reduction of 53.21% and a five-seed range of 35.52--96.00%.

MODEL: Unit cube, circular-average coupling, positive vessel inlet $U=1$; zero exterior tissue potential and sealed other vessel endpoints. $K_3=K_1=1$, radius=0.025, $\kappa=0.05$. Axial flux $q=-K_1\pi r^2 dU/ds$. All quantities dimensionless. Retained disconnected pieces still exchange with tissue.

REGIME: Intact peak tissue potential is less than 0.3% of the imposed inlet value, so feedback on the supplied vessel is weak here. The magnitudes depend on these coefficients and boundary conditions. All test geometries are trees; loops can provide alternative supply paths.

TOY RESULTS:

line | gap_at_0.2 | 68.386 | 76.940 | 80.77

line | gap_at_0.5 | 29.200 | 38.206 | 50.00

line | gap_at_0.8 | 7.674 | 12.254 | 19.23

line | gap_width_0.01 | 28.581 | 37.534 | 50.00

line | gap_width_0.04 | 30.461 | 39.569 | 50.00

y | gap_trunk | 66.432 | 74.833 | 85.64

y | gap_upper | 4.644 | 10.895 | 17.07

y | gap_lower | 4.645 | 10.882 | 17.07

y | gap_both_daughters | 10.527 | 16.151 | 34.66

Columns: geometry, case, inlet flux reduction (%), tissue L2 error (%), detached length (%).

RANDOM BREAKS:

0 breaks: median flux loss 0.000%; range -0.000--0.000%; median tissue error 0.000%.

1 breaks: median flux loss 0.030%; range 0.030--5.391%; median tissue error 0.486%.

2 breaks: median flux loss 0.233%; range 0.060--5.423%; median tissue error 2.297%.

4 breaks: median flux loss 1.535%; range 0.123--11.572%; median tissue error 7.179%.

8 breaks: median flux loss 11.421%; range 5.765--37.576%; median tissue error 21.560%.

16 breaks: median flux loss 18.076%; range 8.213--96.000%; median tissue error 29.382%.

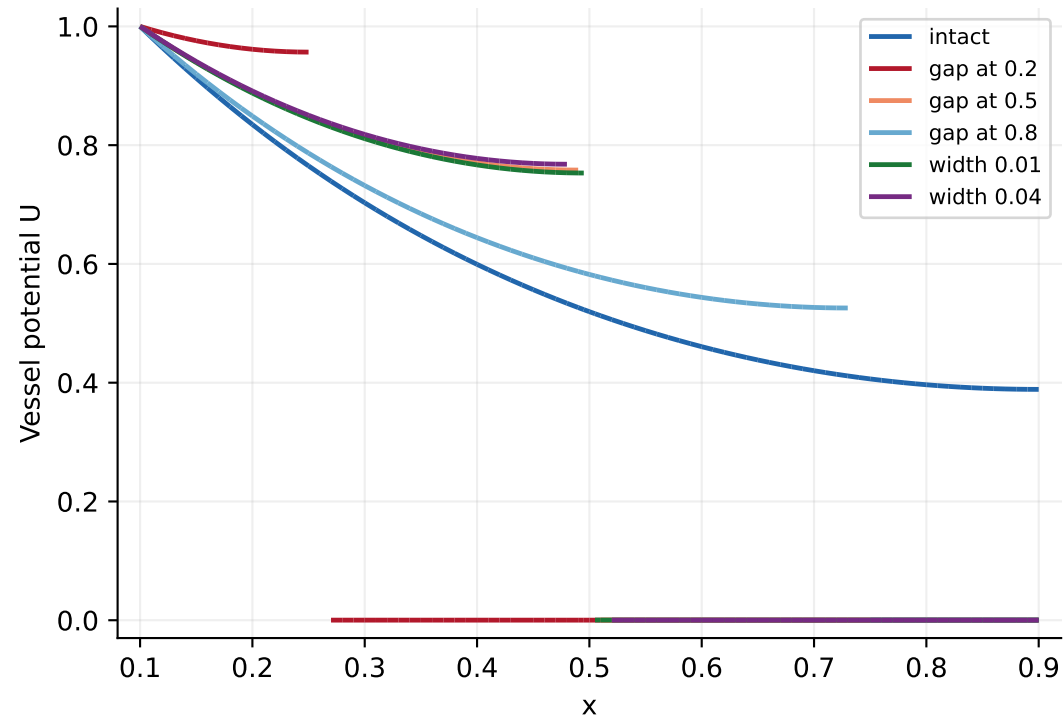
32 breaks: median flux loss 53.211%; range 35.522--96.000%; median tissue error 58.903%.

VALIDATION: Maximum conservation error 8.051e-13; serial/MPI relative difference 1.785e-13; analytic flux order approximately 2.

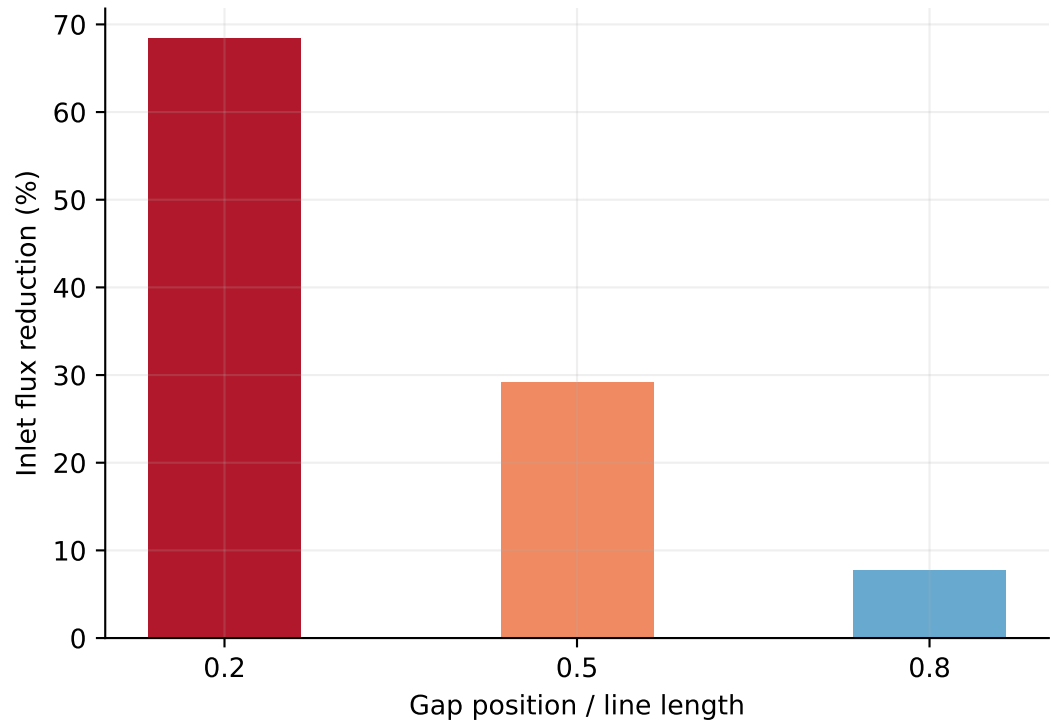
SCOPE: 42 primary synthetic cases. Five random seeds. This is a controlled dimensionless pilot, not a fitted data model. Read REPORT.html or REPORT.md for methods, refinement tables, complete qualifications, and file instructions.

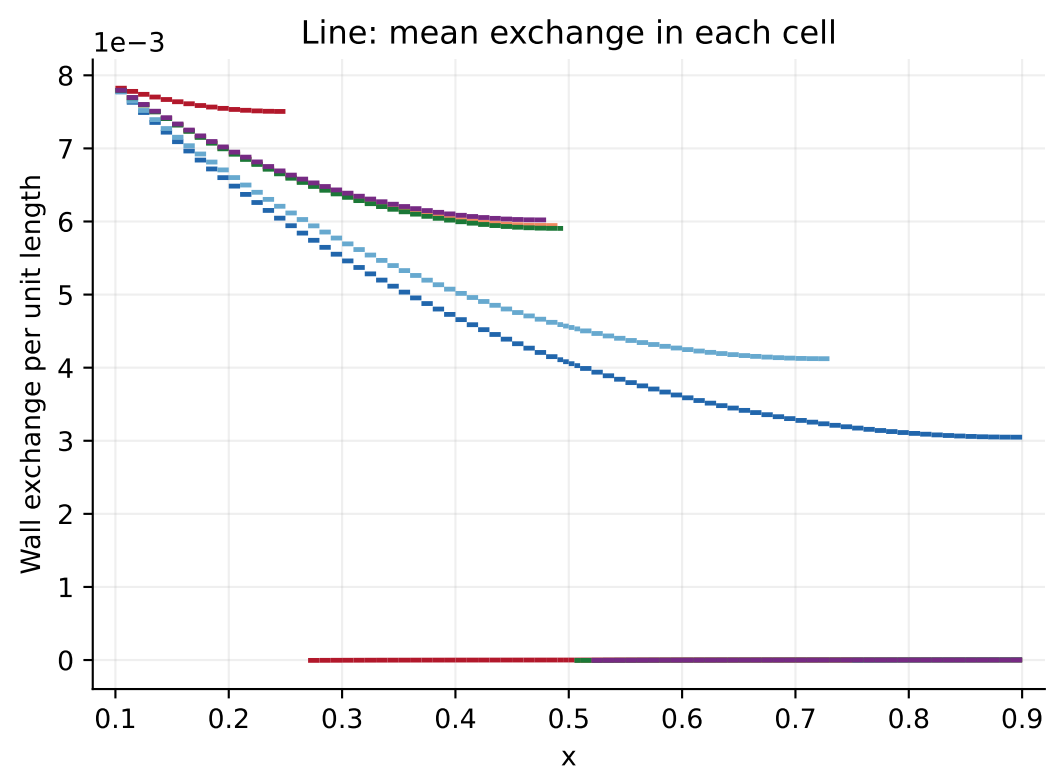
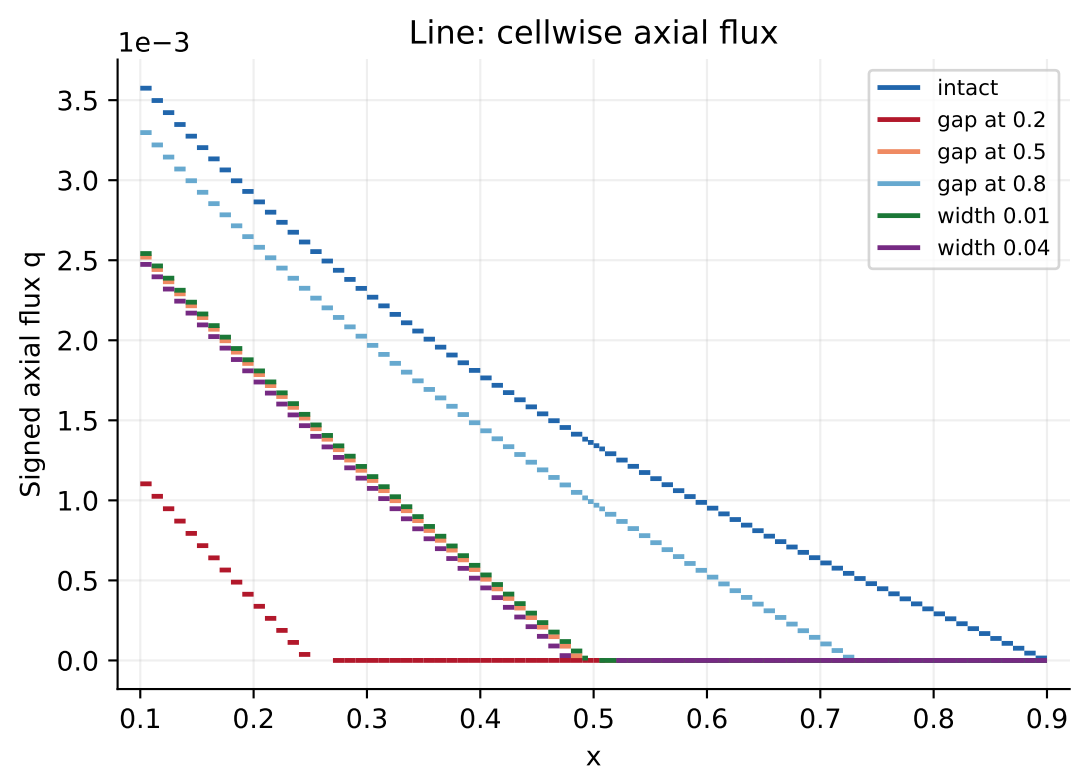
REFERENCE: Laurino and Zunino (2019), DOI 10.1051/m2an/2019042.

Line: pressure on retained segments

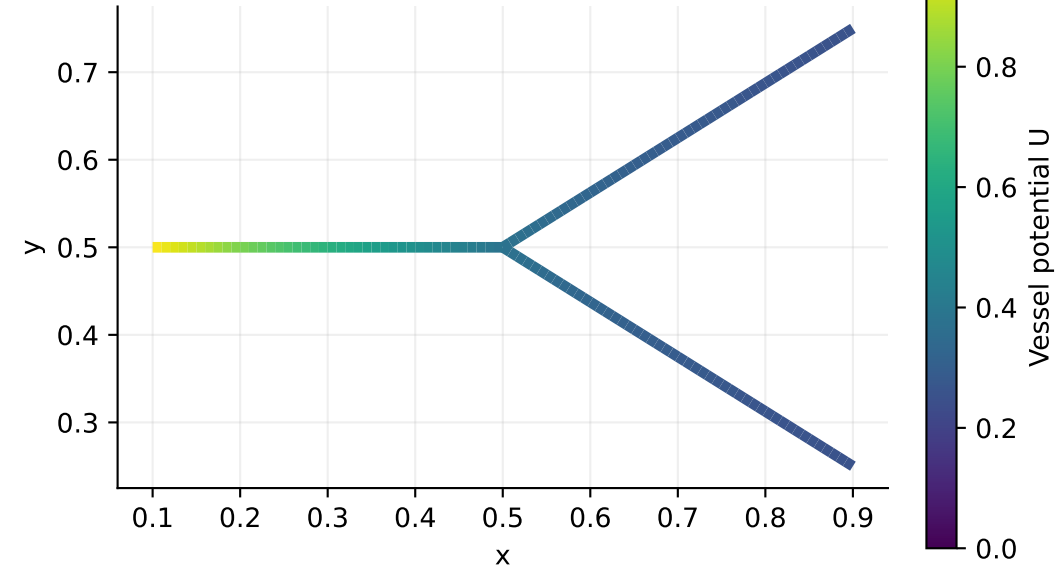


Identical 0.02-long gaps

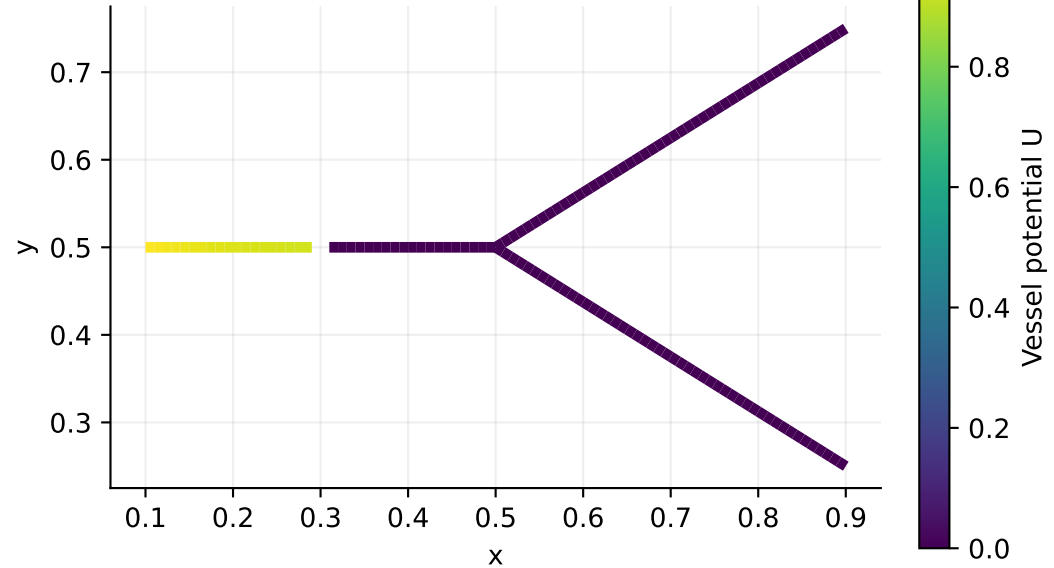




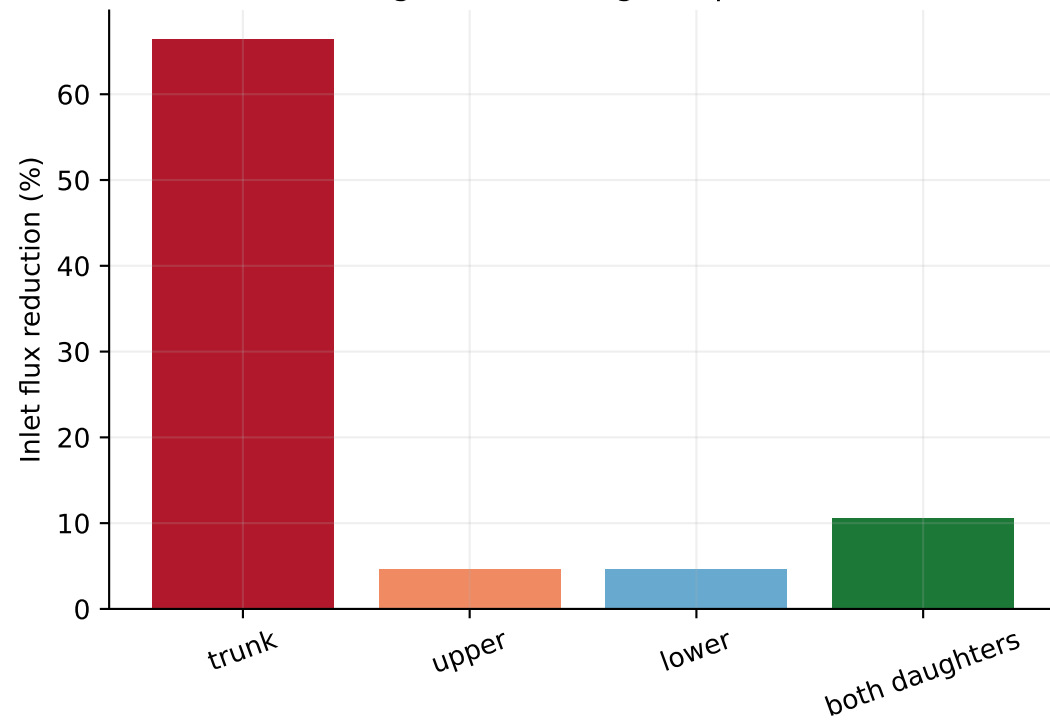
Intact Y



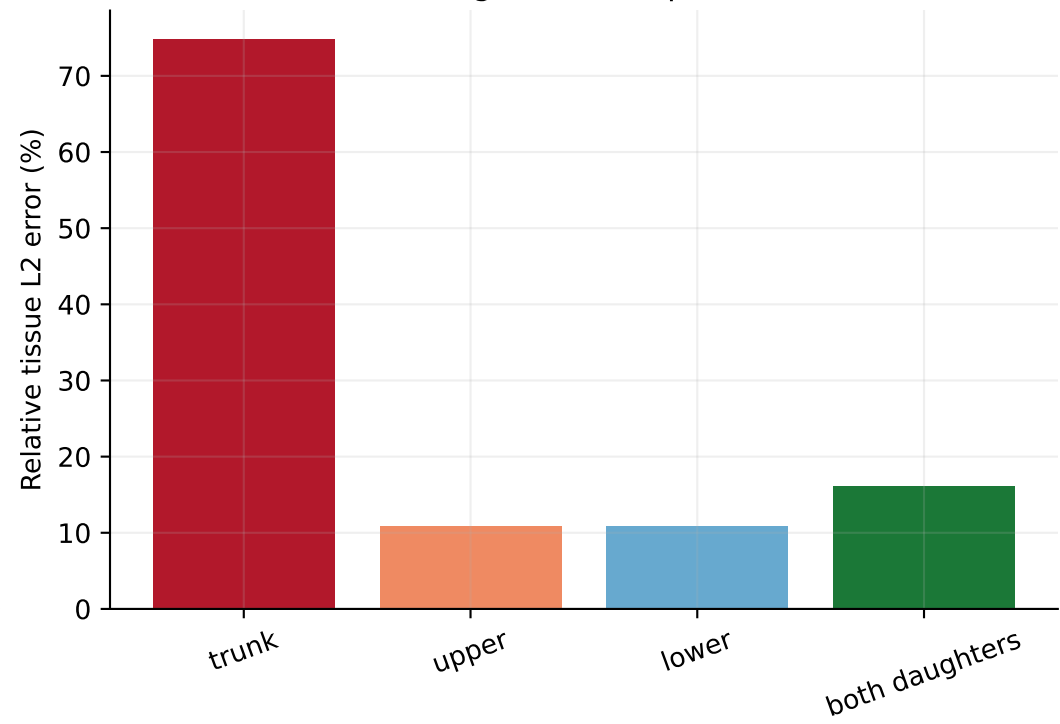
Y with trunk gap

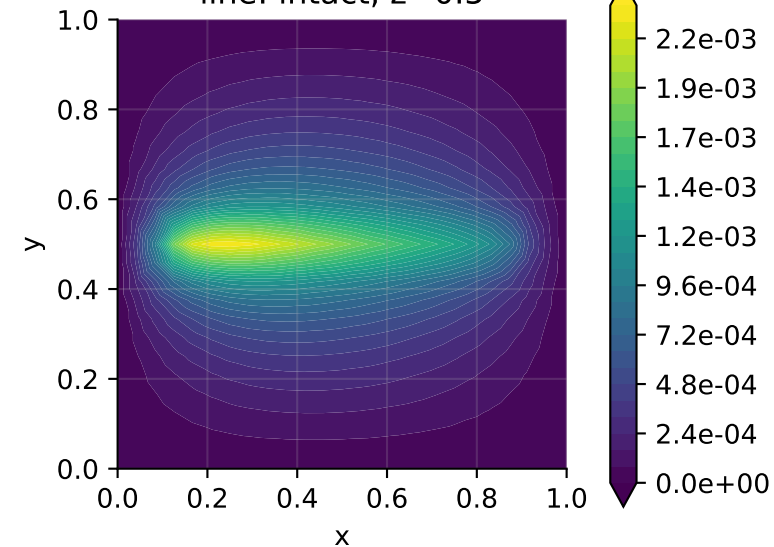
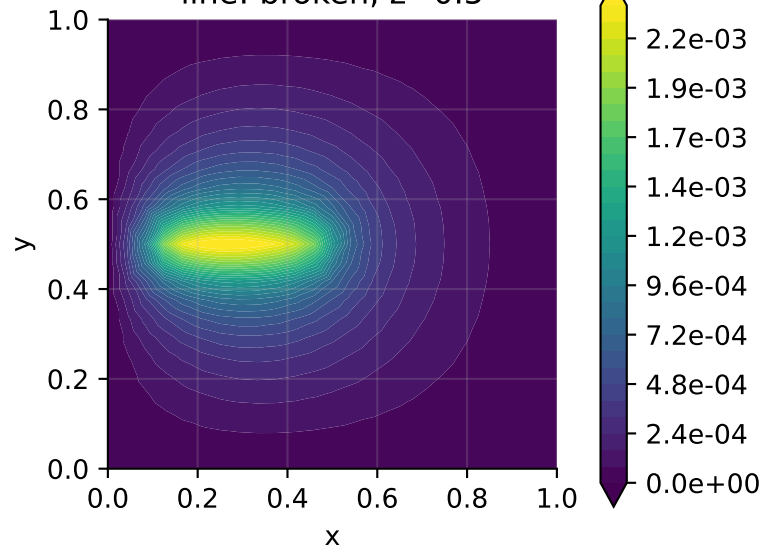
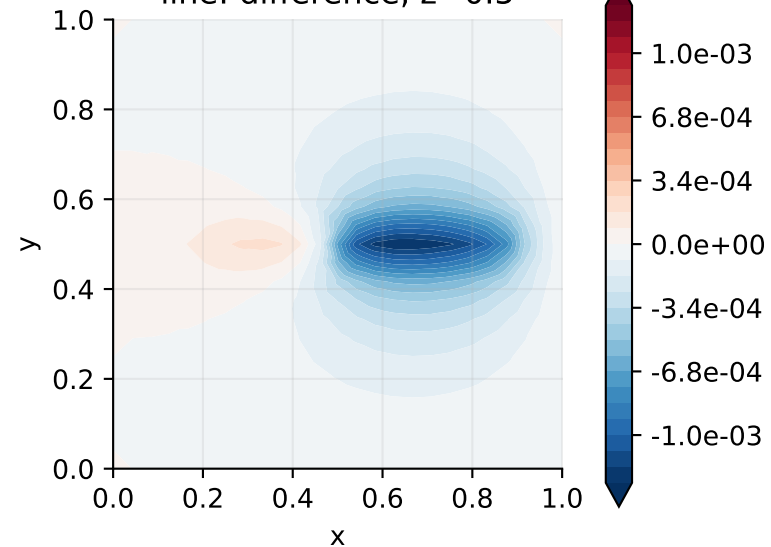
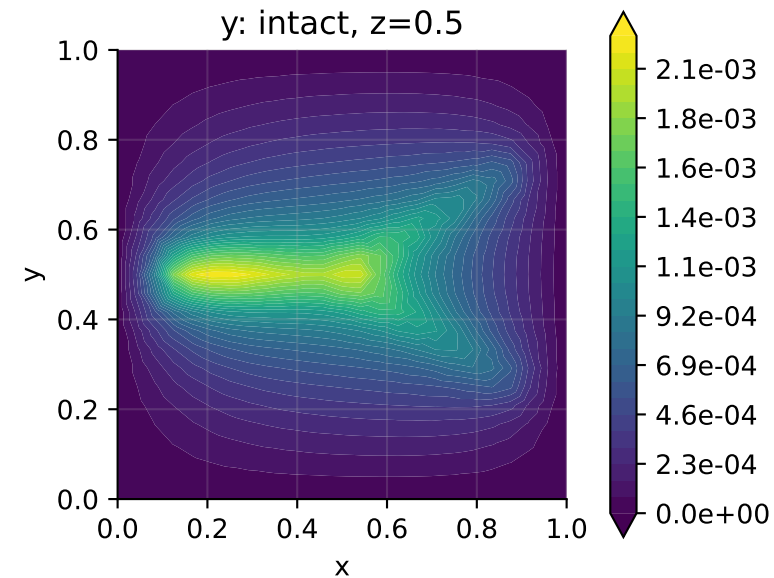
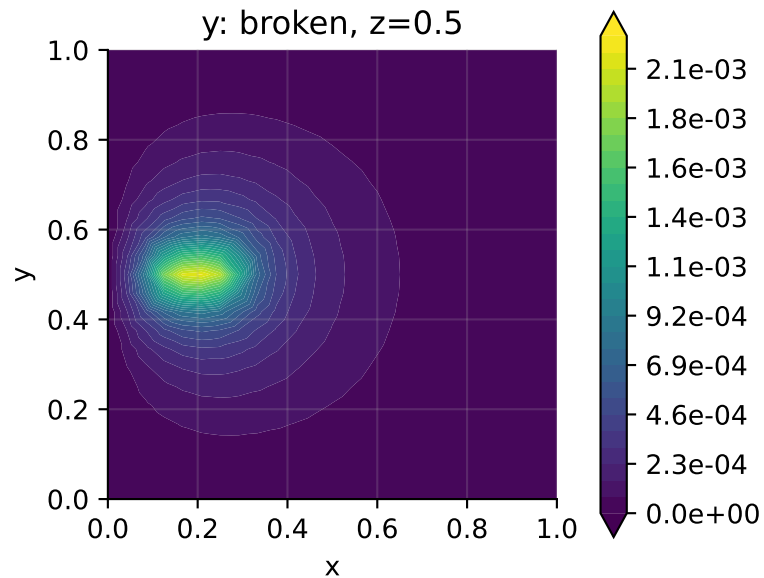
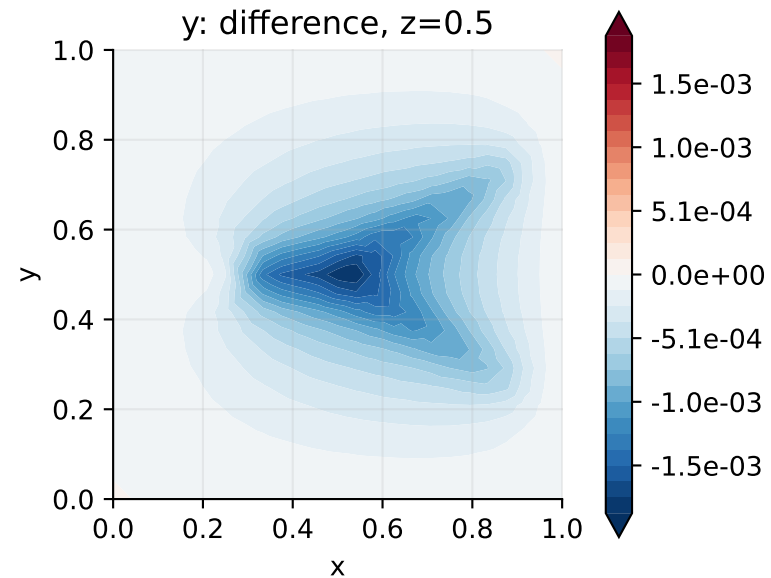


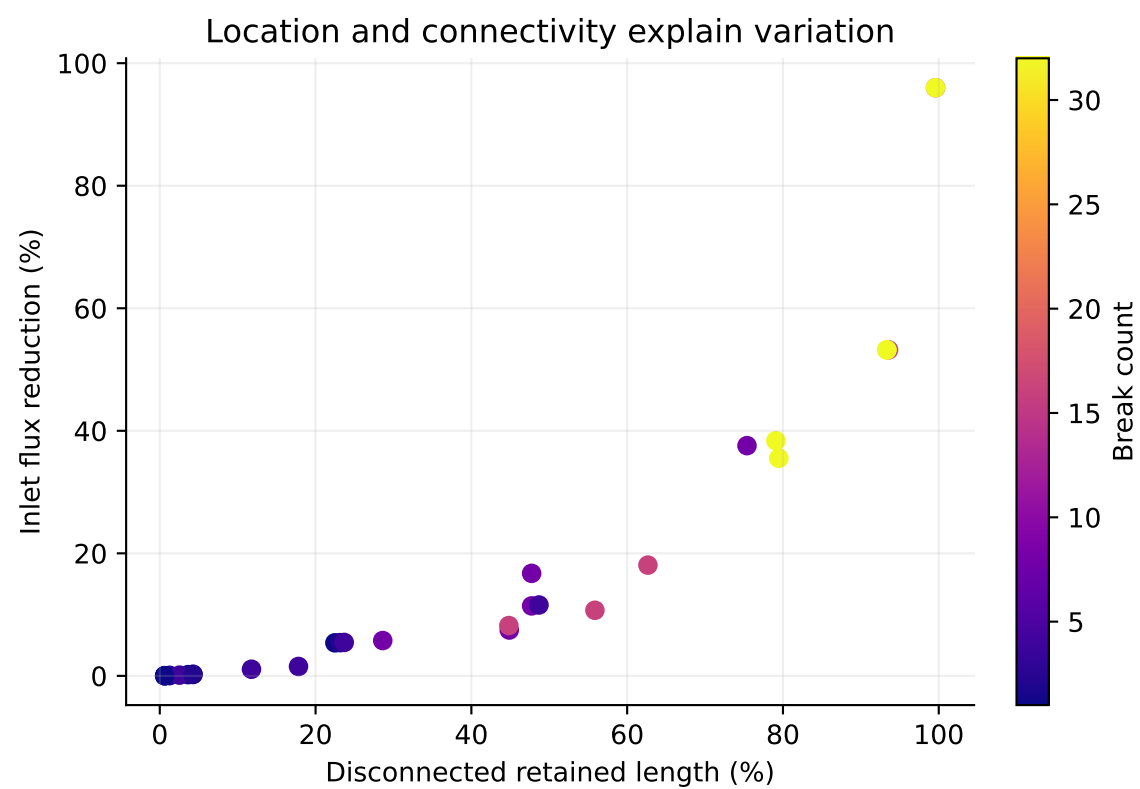
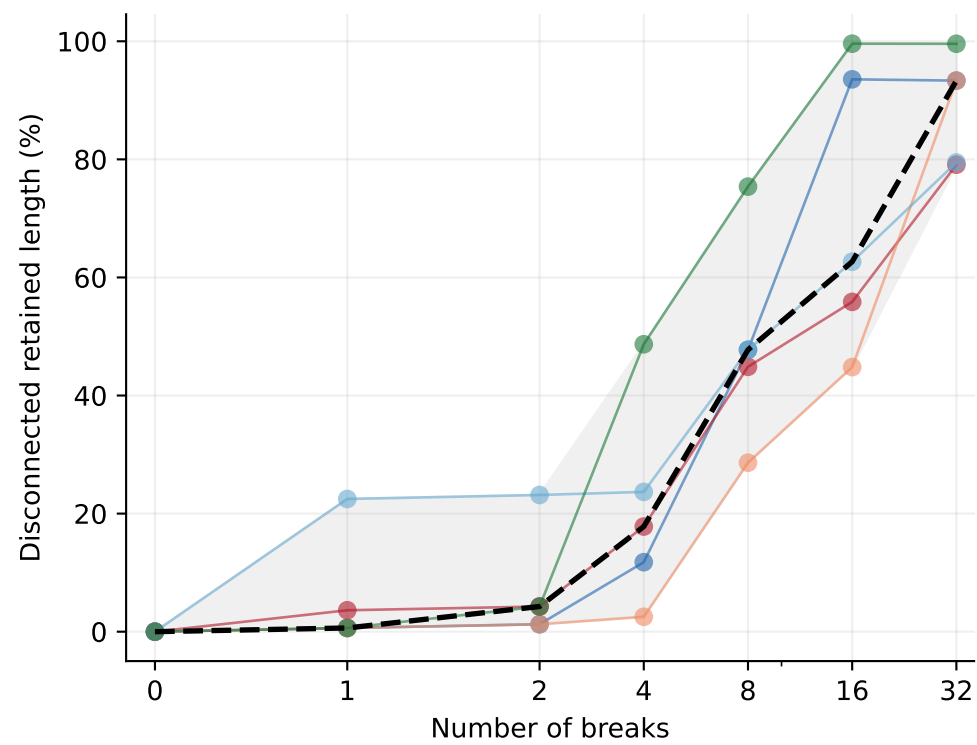
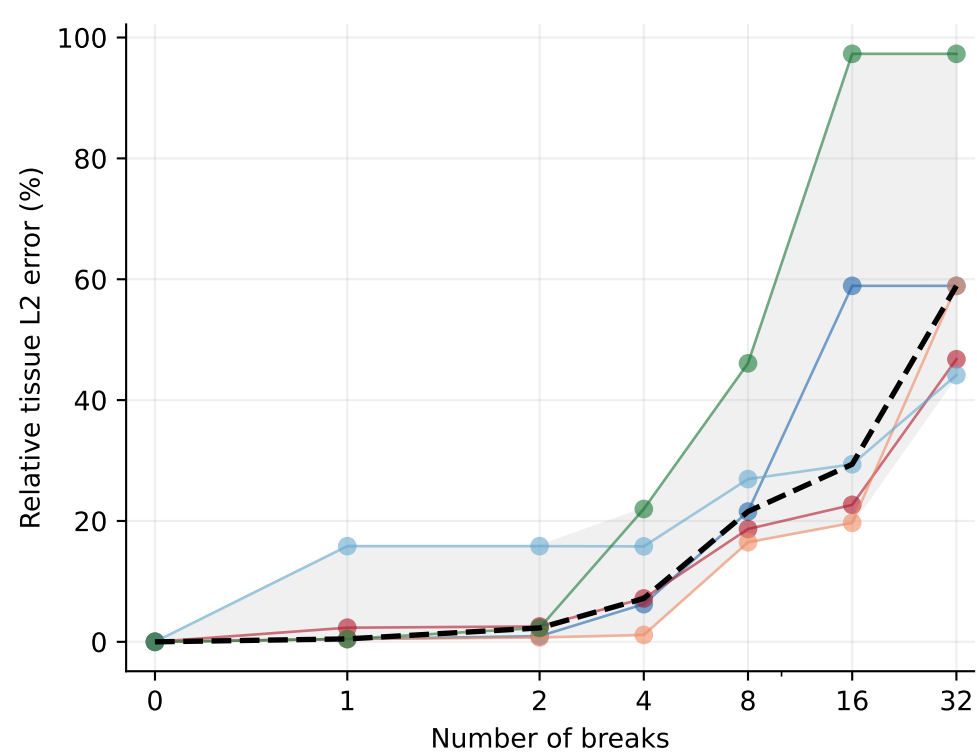
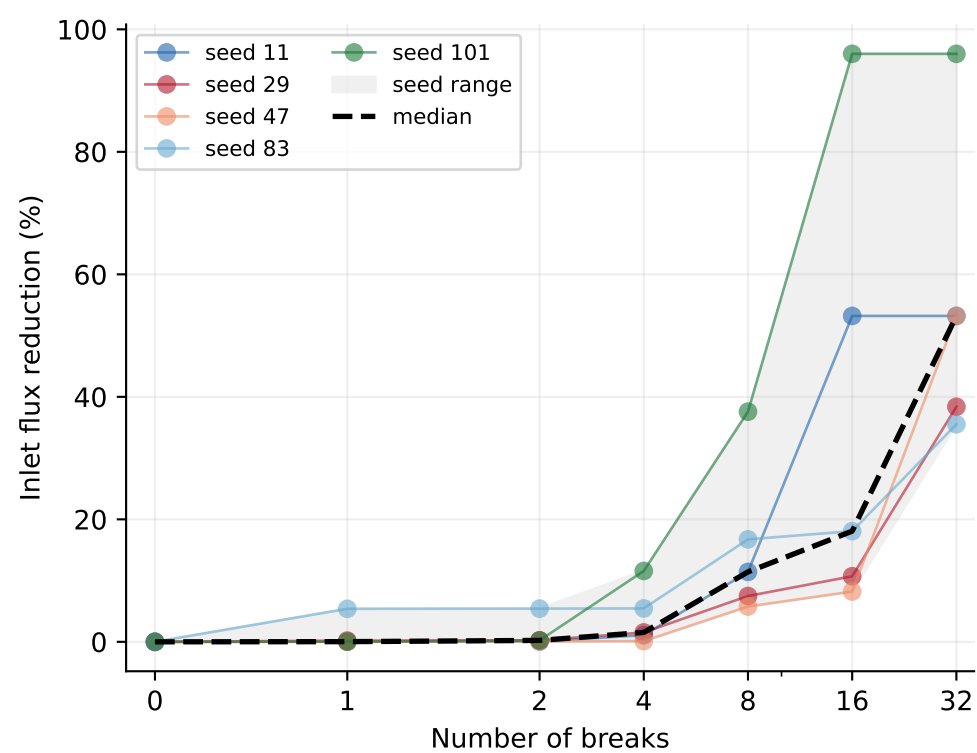
Y: global exchange response



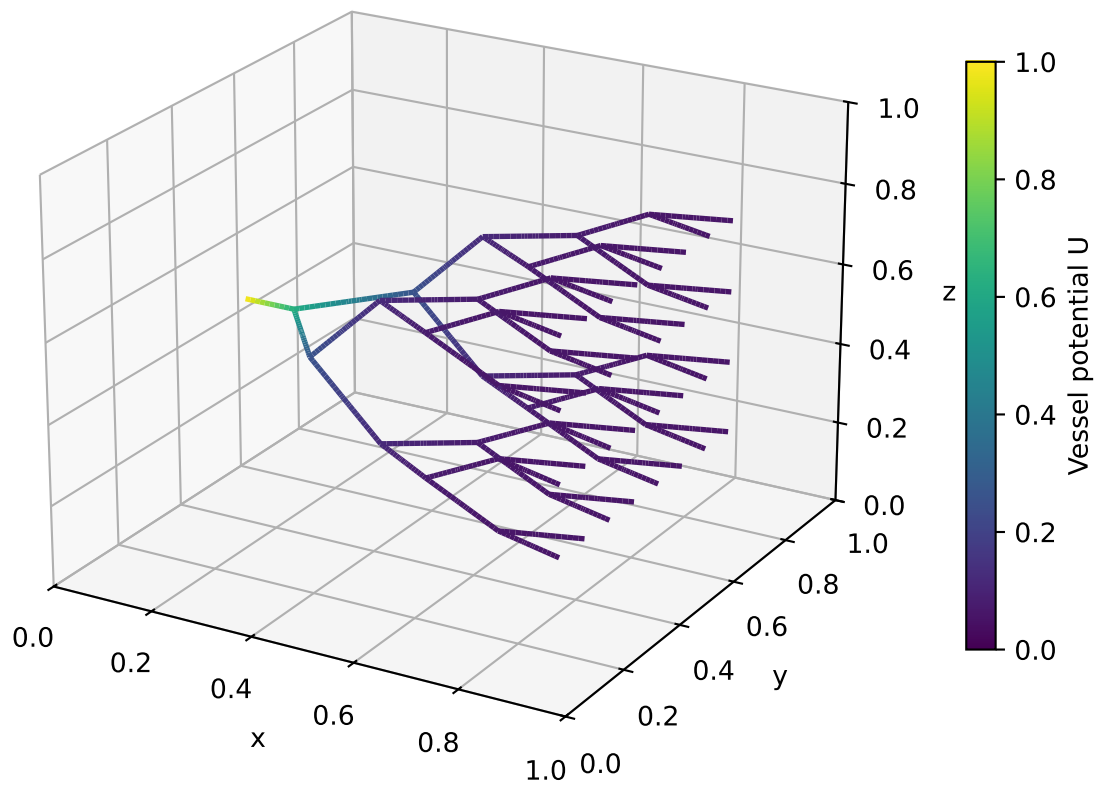
Y: change in tissue potential



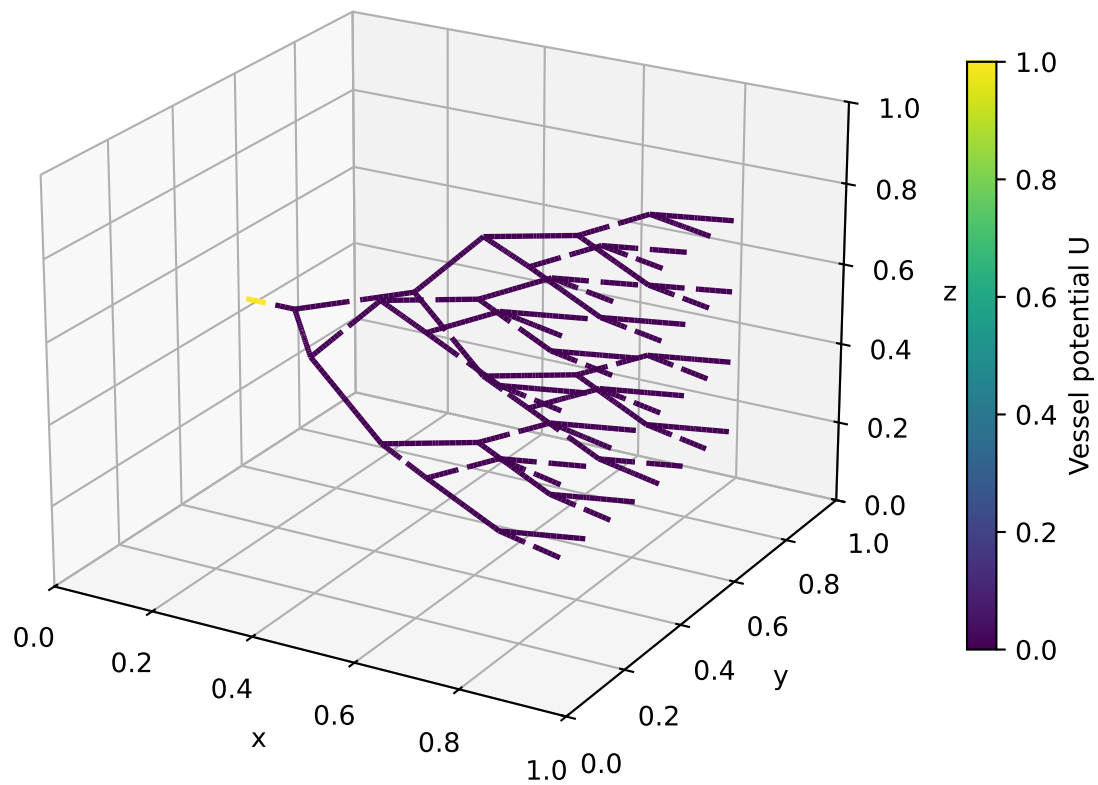
line: intact, $z=0.5$ line: broken, $z=0.5$ line: difference, $z=0.5$ y: intact, $z=0.5$ y: broken, $z=0.5$ y: difference, $z=0.5$ 



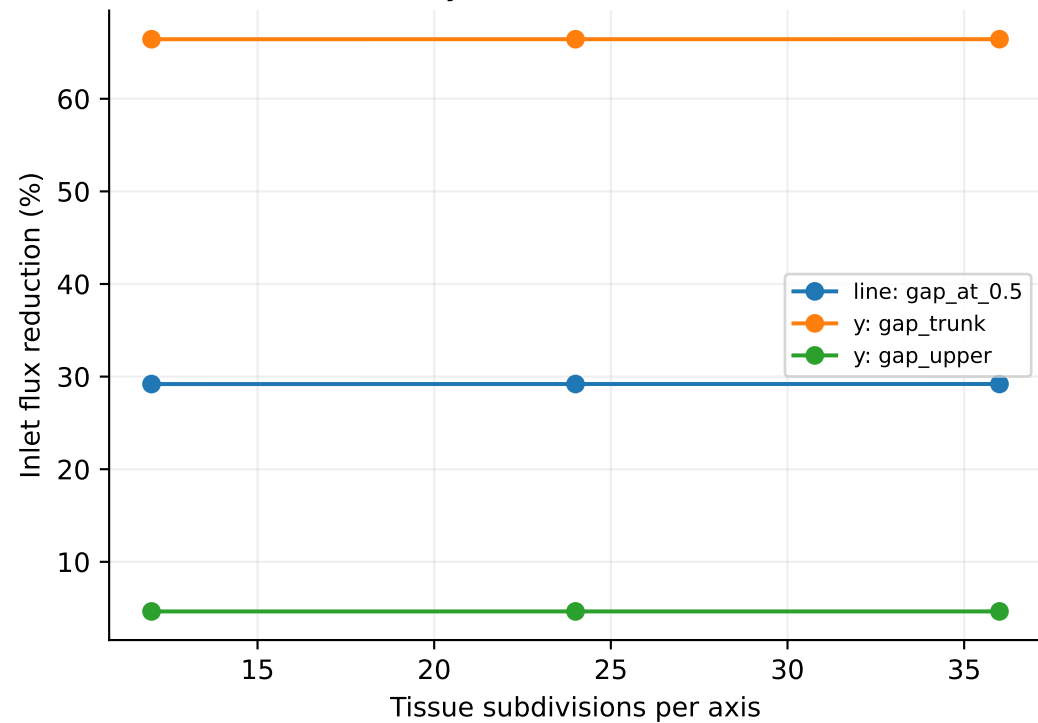
Intact tree



32 breaks, seed 101



Sensitivity versus tissue resolution



Closed-form limit check

